



Wangu Ngalati

Software Engineer

Contact Information:

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Objective

Highly motivated and detail-oriented Software Engineer with a strong background in software development, system integration, and problem-solving. Seeking a challenging position where my skills and expertise in various programming languages and technologies can contribute to the growth of a forward-thinking and innovative company.

Education

DMI St. Eugene University

Bachelor of Computer Science (Software Engineering)

2016 – 2021

- Graduated with a GPA above 4.0
- Awarded top student honors from 2017 to 2019

David Kaunda National Technical High School

2012 – 2014

- Full Grade 12 Certificate

Technical Skills

Programming Languages:

Java, JavaScript, Kotlin, Dart, Python, PHP, HTML, CSS, C++, C, SQL, Golang

Frameworks & Libraries:

JQuery, Android Studio, Flutter, Laravel, Yii, Django, Node.js (Express), React, Bootstrap, Material UI, CSS, SpringBoot

Databases:

MySQL, Firebase (Firestore, Realtime Database), MongoDB, Postgres, MariaDB

DevOps & Containerization:

Docker, Docker Compose

Cloud & Deployment:

AWS Deployment (EC2, S3), Google Cloud Platform (GCP), c-panel, WHM (Web Host Manager)

Google APIs Intergration:

Messaging API, Directions API, Routing API, Geolocation API

Other Skills:

- UI/UX Design: Adobe XD, JustInMind
- Game Development: Unreal Engine
- Graphics Design: Photopea, CorelDRAW, Photoshop
- 3D Animation: Blender
- Containers: Docker
- Runtime Environments: Node.js

Professional Experience

EMU Technologies Limited (Backend Developer – Ongoing)

May 2025 – Present

- Developed and managed multiple client websites, including: fiatzambia.com, abaleyadriving.com, quartanmining.com, dropshipltd.com, semassmetals.com, sophonzambia.com, komelazm.com, copperpotzm.com, kutwilamedia.com, and abaleya.name.
- Built and deployed emuzm.com, a reseller platform providing web hosting services, including domain and IP binding, cPanel/WHM configuration, and email integration for Outlook.
- Designed and implemented a daily quote system delivering motivational, health, and fitness messages to users via automated scheduling.
- Managed backend development in PHP and Laravel, including database modifications and updates on MySQL hosted on AWS EC2.

- Configured and monitored system alerts and SMS notifications for the Abaleya carpooling platform to notify management of new users, vehicle registrations, and trip approvals.
- Implemented in-app notifications and approval alerts to improve communication between the platform and users.
- Integrated Agora live streaming services, including real-time video streaming features.
- Implemented WebSocket-based functionality for live view counts and real-time comments on live streams.
- Configured and utilized Cloudflare services, including Cloudflare Stream and Cloudflare Images for media delivery and optimization.
- Implemented adaptive bitrate streaming (ABS) using AWS S3 and MediaConvert, as well as Cloudflare, enabling seamless video quality progression from 360p to 1080p.
- Managed DNS configurations, including A record updates and domain transfers.
- Configured and supported Outlook email accounts for clients and internal use.
- Performed computer installation, maintenance, and servicing (hardware and software).
- Performed system testing, IT support, and hardware/software servicing to ensure smooth operations across the company.

Abaleya Carpooling (Backend Developer – Contract)

November 2024 – May 2025

- Developed the complete backend architecture for the Abaleya ride-sharing platform, ensuring scalable and maintainable infrastructure.
- Integrated Zynle Pay for secure and seamless in-app payment processing.
- Integrated SMSZambia.com to enable real-time SMS notifications for ride confirmations, alerts, and user communication.
- Implemented Firebase Cloud Messaging to deliver real-time in-app notifications for trip updates and critical alerts.
- Integrated Google's Places, Directions, and Geolocation APIs to enable accurate location tracking, trip routing, and address suggestions.
- Integrated SendMail API for email communications, including verification, trip receipts, and service updates.
- Built a secure registration system with OTP (One-Time Password) verification to validate user authenticity during sign-up.
- Designed and implemented a referral system, including a dashboard to monitor referral performance—highlighting users with the highest and lowest scores and identifying who referred whom.
- Developed an SOS alert system for passengers to discreetly notify the platform in case they felt unsafe during a ride, enhancing rider security and trust.

Chrilan Technologies (Backend Developer - Contract)

June 2024 – November 2024

- Designed and implemented batch files to automate processes, such as activating printing services on Point-of-Sale (POS) computers running Lysoft software.
- Developed VAT report API endpoints for Yango, streamlining financial reporting processes.

- Created OTP generation endpoints with secure session management stored in the database for enhanced user authentication.
- Built APIs to adjust barcodes for smart invoicing, improving accuracy and efficiency in invoicing workflows.
- Developed JavaScript interfaces to enable seamless interaction with Lysoft's mobile printing service from the web version of their application.
- Contributed to system optimization and integration efforts, ensuring smooth functionality across various services and platforms.

PrimeNet Solutions (Backend Developer - Contract)

February 2024 – March 2024

- Integrated SMS and payment gateway systems
- Developed a ticketing system using Twak.io and Freshdesk
- Conducted API testing for TransUnion and DotGov for clients like ZamPost eWallet and Panda Africa
- Optimized loan data retrieval using TransUnion API, reducing costs and improving response times
- Enhanced UI and performed server-side optimization for the PrimeNet SMS and Payment Gateway

Frontier Finance (Software Developer & Loan Credit Officer - Contract)

October 2023 – February 2024

- Developed a loan status widget to track collections
- Built a dynamic PDF generator for loan agreements
- Automated Google Drive folder creation for loan applicants, integrating detailed user data

Digital Treasure Zambia (Software Developer - Full-time)

July 2022 – April 2023

- Designed and developed the Early Learning Kiosk File Manager and ELK Launcher for the Roger Federer Foundation
- Created background services for the Siyafunda Zulu language learning app using Kotlin
- Developed a beauty appointment booking system (Beta Looks) using Kotlin Multiplatform
- Integrated a JSON file importer and created modules for Smart Forms and Crime Scene reporting using Firebase

Texila American University (IT Specialist & Network Engineer Assistant - Internship)

August 2021 – December 2021

- Set up Microsoft Teams, installed and serviced computers, and configured domain networks with NComputing

Blue Code Systems Limited (Software Developer - Internship)

June 2019 – December 2019

- Queried data from OpenMRS and designed the SmartCerv application
- Prepared DHIS2 reports using ETL Python scripts

Zambia Broadcasting Corporation (IT Specialist & Network Engineer Assistant - Internship)

June 2018 – December 2018

- Managed static IP configurations, network connections, and Ethernet cabling

- Installed and serviced computers for network connectivity

Notable Projects

Advanced Calculator & ATM Simulator

- Simulates ATM transactions and performs complex calculations -
- Technologies: Java, MySQL

Nayenda Bus Ticketing System

- Real-time ticketing application for traveling customers
- Technologies: Java, Firebase (Authentication, Storage, Realtime Database)

Ubulimi (Agro-Advisory Application)

- Provides crop decision-making tools and weather predictions for farmers -
- Technologies: Kotlin, Firebase

Vitundu Language App

- Teaches Zambian children local languages through an interactive learning platform -
- Technologies: Kotlin, Firebase

Murder Hornet Character Design

- Designed a character for ProSam Studio animations
- Technologies: Krita, Huion Tablet, Blender

Graphics & Logo Design

- Designed logos, posters, brochures, and letterheads for various companies
- Technologies: Photopea, CorelDRAW, Photoshop

References

Mr. Choko

Human Resource Manager

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Chief Software Architect

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Riteware Technology, Lusaka, Zambia